

AI-assisted course workflow

CP4285 · MODERN RECOMMENDATION SYSTEMS

AI drafts, compiles, and interrogates course materials. The instruction team reviews every teaching and assessment decision.

Slide generation: a renderable teaching artifact

Goal: turn a teaching brief into an inspectable, versioned lecture deck without treating generated prose as course truth.

1. Ground the brief

Read the topic, textbook, learning outcomes, prior slides, and course constraints. Record sources and activity requirements.

2. Generate structured source

Draft `.qmd` with shared CP4285 primitives, speaker notes, links, alt text, and explicit claims.

3. Compile and inspect

Run Quarto at 1600 × 900. Check every state for text fit, contrast, broken assets, source accuracy, and delivery timing.

Instructor gate: approve the learning sequence, factual claims, examples, and assessment implications before the rendered `.html` becomes lecture material.

Traceability records the prompt, source files, Git diff, rendered deck, and visual review.



Website generation: source, build, deploy

Goal: publish supporting course information from version-controlled content while preserving the slides as the definitive course record.

1. Edit the source layer

Update Markdown in `content/`, Hugo configuration, and reusable layouts. Keep dates, links, policy text, and navigation explicit.

2. Build the static site

Hugo resolves templates and content into `public/`; Pagefind builds the search index from the generated pages.

3. Test the published surface

Inspect local output for path correctness, responsive layout, accessible names, stale dates, and consistency with the lecture slides.

Instructor gate: confirm that public pages communicate the approved policy and link to the correct course materials. Generated output never replaces source review.

Traceability records content, templates, the Hugo build, generated `public/`, and deployment review.



Quiz generation: candidate questions under human control

Goal: use AI to help form a question bank, then validate every item as an assessment rather than a plausible-looking prompt.

1. Specify the construct

Start from a learning outcome and a bounded lecture section. State the intended reasoning, format, difficulty, and evidence students may use.

2. Produce candidates

Generate questions with answer keys, distractor rationales, calculations or worked solutions, and references to the relevant course material.

3. Validate before release

Min checks correctness, ambiguity, accessibility, difficulty, duplication, and alignment. Peer suggestions remain candidates until curation.

Instructor gate: publish only reviewed items with a defensible answer and rubric. In-class quizzes remain AI-free, even though AI may support their preparation.

Traceability links the CLO and course evidence to the candidate bank, instructor review, and controlled release.



Voice agents: Socratic discussion with a visible control loop

Goal: use a voice agent to surface reasoning during a discussion, then let students and the instructor evaluate the reasoning rather than defer to the voice.

1. Scope the dialogue

Provide a bounded case, the relevant course concepts, a question sequence, and clear limits. Exclude personal data and grading decisions.

2. Run the question loop

Speech recognition captures a student claim. The agent tracks the current turn, asks for evidence or a counterexample, then speaks the next question.

3. Debrief the record

Use the transcript to identify unsupported assumptions, missing stakeholders, and competing explanations. Students challenge the agent's reasoning.

Instructor gate: choose the case, stop or redirect an unhelpful exchange, and close with the course-grounded synthesis. A voice agent facilitates a discussion. It does not adjudicate it.

Traceability records the scoped prompt, source material, transcript, student critique, and instructor synthesis.



Canvas inputs: AI-assisted entry, instructor-approved publication

Goal: reduce mechanical entry work in Canvas while keeping course policy, audience, and publication decisions with the instruction team.

1. Form an approved payload

Start from approved dates, learning outcomes, policy text, links, and audience. Represent each field explicitly: title, body, deadline, availability, and attachments.

2. Populate a draft

AI can format announcements, discussion prompts, assignment instructions, or quiz metadata for the corresponding Canvas fields and flag missing information.

3. Preview and reconcile

Check the rendered Canvas page, links, student view, recipients, points, dates, availability windows, and consistency with the slides and website.

Instructor gate: a staff member reviews the exact Canvas form and explicitly publishes it. AI may prepare inputs; it does not independently release student-facing changes.

Traceability records the approved source, structured Canvas draft, staff preview, and explicit publication approval.

